[Binary classification]

| x | y | z | a | output |
|---|---|---|---|---|
| DSA | Attendance | Project | Aptitude | |
| 70 | 78 | 3 | 70 | 0 |
| 80 | 76 | 4 | 80 | 1 |

0.7 ⇒ 70%

$$\text{Output} = [w_1 x + w_2 * y + w_3 * z + w_4 * a + \theta]$$

$w_1 = 3$ $= \boxed{80}$

$w_2 = 4$

$w_3 = 0.2$

$w_4 = -5$

$\theta = 10$

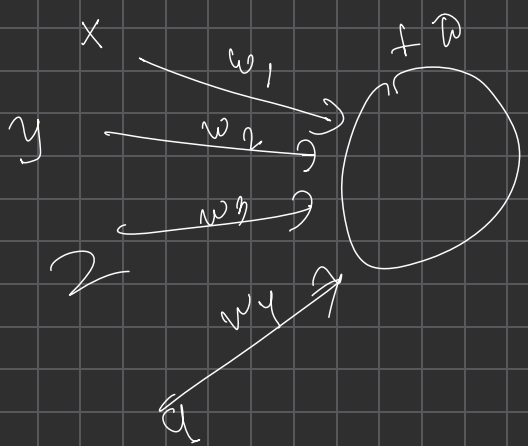
$\boxed{0.7}$ → real output → 1

cost

* Loss function ✓ $1 - 0.7$

$= \boxed{0.3}$

$w_1 = w_1 \text{old} + 0.01 \times 0.3 \times 20$



Output = 90 $\Rightarrow$ 0

$\downarrow$

0.8

Error = 0 - 0.8

$\Rightarrow$ = -0.8

$w_1 = w_{10} + 0.01 \times -0.8 \times$ Input

Threshold

$\downarrow$

270

$\rightarrow$ X = 10 lakh $\Rightarrow$ 0.99995 - - -

$\rightarrow$ X = 10K $\Rightarrow$ 0.99914

Error = 0.0006

$\Rightarrow$ 1

X = Output $\Rightarrow$ 0.8

X $\Rightarrow$ Output $\Rightarrow$ 0.3 $\Rightarrow$